Problem 1 (1 pt)**.** Recall that a *directed graph* (or digraph) is a pair $D = (V, E)$ where V is a set and $E \subseteq \{(u, v) \in V \times V : u \neq v\}$. For $u, v \in V$, a u-v diwalk in D is a sequence $(u = v_0, \ldots, v_k = v)$ such that $(v_i, v_{i+1}) \in E$ for all $i \in \{0, \ldots, k-1\}$.

Consider the relation R on V where $u \,R\, v$ iff there is a u-v diwalk. Give an example of a digraph D where R is **not** an equivalence relation. Justify your answer. (Feel free to draw a picture of D)

Solution. Consider the digraph D with two vertices $\{a, b\}$ and a unique arc $E = \{(a, b)\}$. Then $a \,R\, b$, because (a, b) is an a–b diwalk. However, since (b, a) is not an arc in D, it follows that there is no b–a diwalk. Thus b is **not** related to a, and the relation R fails to be symmetric. □

Problem 2 (2 pts)**.** Let G be a connected graph and consider a function $f\colon V(G) \to X$ where X is some arbitrary set. Prove that if f is *not* a constant function, then there is an edge $uv \in E(G)$ such that $f(u) \neq f(v)$.

Solution 1. Let us show the contrapositive statement. Thus we assume that for any pair of vertices u, v in G, if $uv \in E(G)$ then $f(u) = f(v)$. To see that f must be constant, fix any vertex u^* in G. Now, for any other vertex v, since G is connected, there is a u^*–v path in G, say $(u^* = v_0, v_1, \ldots, v_n = v)$. This means that $v_i v_{i+1} \in E(G)$ for each $i \in \{0, 1, \ldots, n-1\}$, thus $f(v_i) = f(v_{i+1})$ for any such i. Putting these n-equalities together, we get

$$f(u^*) = f(v_0) = f(v_1) = f(v_2) = \cdots = f(v_{n-1}) = f(v_n) = f(v).$$

Thus $f(v) = f(u^*)$ for any $v \in V(G)$, and f is constant. □

Solution 2. Since f is not constant, there are vertices u, v such that $f(u) \neq f(v)$. Since G is connected, there is a u–v path in G, say $(u = v_0, v_1, \ldots, v_n = v)$. We claim that there must be some $i \in \{0, 1, \ldots, n-1\}$ such that $f(v_i) \neq f(v_{i+1})$. Indeed, otherwise

$$f(u) = f(v_0) = f(v_1) = f(v_2) = \cdots = f(v_{n-1}) = f(v_n) = f(v),$$

which contradicts $f(u) \neq f(v)$. Thus there is some $i \in \{0, 1, \ldots, n-1\}$ such that $f(v_i) \neq f(v_{i+1})$, and we certainly have $v_i v_{i+1} \in E(G)$ because $(u = v_0, v_1, \ldots, v_n = v)$ is a path.

Problem 3 (2 pts)**.** Prove that every graph on at least two vertices has a pair of vertices with the same degree.

Solution. Suppose G has n vertices, thus $\deg v \in \{0, \ldots, n-1\}$ for each $v \in V(G)$. Suppose for a contradiction that no two vertices have the same degree. In this case we can label the vertices $v_0, v_1, \ldots v_{n-1}$ so that $\deg v_i = i$ for each i. Since v_{n-1} has degree $n-1$, it must be adjacent to any other vertex, in particular to v_0. This contradicts that $\deg v_0 = 0$. □

Problem 4 (2 pts)**.** Prove that if G is a graph with $\delta(G) \geqslant 2$, then G must contain a cycle.

Solution. Take a **maximal** path $P = (v_0, v_1, \ldots, v_k)$. Since $\delta(G) \geqslant 2$, we have in particular $\delta(v_0) \geqslant 2$, which means that there is some $u \in V(G)$ with $u \neq v_1$, such that $v_0 u \in E(G)$. Now we claim that $u \in V(P)$. Indeed, if not, then $(u, v_0, v_1, \ldots, v_k)$ is a path in G, contradicting the maximality of the path P. Hence $u = v_i$ for some $i \geqslant 2$, because $u \neq v_1$. This means that $v_i v_0 \in E(G)$ for some $i \geqslant 2$, therefore $(v_0, v_1, \ldots, v_i, v_0)$ is a cycle in G. □

Problem 5 (3 pts)**.** Let G be a graph and let A be an independent set of G. Prove that

$$\sum_{v \in A} \deg v \leqslant |E(G)|$$

with equality if and only if G is bipartite with parts A and $V(G)\backslash A$.
(Especially in this problem, be sure to carefully justify all steps in your argument)
Hint: consider the sets $E_v = \{\text{edges incident with } v\}$ and $\widehat{E} = \bigcup_{v \in A} E_v$. For the equality, decide when is $\widehat{E} = E(G)$.

Solution. We need to show the following

(1) $\sum_{v \in A} \deg v \leqslant |E(G)|$, and

(2) $\sum_{v \in A} \deg v = |E(G)|$ if and only if G is bipartite with parts A and $V(G)\backslash A$.

Following the hint, consider the sets $E_v = \{\text{edges incident with } v\}$ (for $v \in A$) and $\widehat{E} = \bigcup_{v \in A} E_v$.

To show (1), we first claim that for any pair of different vertices u, v in A, we have $E_u \cap E_v = \varnothing$. Indeed, if $e \in E_u \cap E_v$, then $u \in e$ and $v \in e$. But since $u \neq v$, we get $e = uv$. This is impossible, since A is independent. Since $\widehat{E} \subseteq E(G)$, and using the previous claim, we get

$$|E(G)| \geqslant |\widehat{E}| = \sum_{v \in A} |E_v| = \sum_{v \in A} \deg v.$$

For part (2), notice first that since $V(G) = A \sqcup (V(G)\backslash A)$ is a partition, and since A is independent, it follows that $V(G) = A \sqcup (V(G)\backslash A)$ is a bipartition if and only if $V(G)\backslash A$ is an independent set. Thus, by the previous argument, to prove (2) we need to show that $E(G) = \widehat{E}$ if and only if $V(G)\backslash A$ is independent.

Assume first $E(G) = \widehat{E}$; thus any edge e in G is actually in $\widehat{E}$, which means that e is incident with a vertex in A. Hence G contains no edges between vertices in $V(G)\backslash A$, which means that $V(G)\backslash A$ is independent. Conversely, if $V(G)\backslash A$ is independent, each $e \in E(G)$ contains a vertex in A, which means that $e \in \widehat{E}$; thus $E(G) = \widehat{E}$. □